

Demonstration of Proposed Features

OptiCrop — Smart Agricultural Production Optimization Engine

Feature 1: Home Page

Demonstrates project introduction and navigation. Shown by loading the application root URL and pointing out the navbar links to About and Find Your Crop.

Feature 2: About Page

Demonstrates project context — dataset, methodology, and model performance (Random Forest, 99.55% accuracy) — shown by navigating to /about.

Feature 3: Find Your Crop (Core Feature)

Demonstrates the core ML-powered recommendation: a known soil/climate profile (e.g. N=90, P=42, K=43, temperature=20.88, humidity=82.0, pH=6.5, rainfall=202.9) is entered, and the system correctly returns 'Rice' — matching the actual dataset label for that profile, demonstrating the model is making genuine, verifiable predictions rather than arbitrary output.

Feature 4: Model Comparison (Notebook)

Demonstrates the underlying rigor — the training notebook shows 4 algorithms compared side-by-side on identical train/test splits, with Random Forest selected transparently based on superior metrics rather than an arbitrary choice.